ANIMAL WELFARE & ECOSYSTEM RESTORATION

# Tierra del Fuego

A restoration decision-support platform for the Culebra wildfire

## Group N ANWE




Product image supplied by Group N ANWE

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**DEMO CASE**

**Sierra de la Culebra, Spain**

**EVIDENCE SOURCES**

**CEMS, Sentinel-2 and GBIF**

**DECISION STATUS**

**Unreviewed draft / field checks required**

01. Relevance

# Restoration decisions after wildfire

June 2022 Culebra case
Animal needs remain a field question

## The decision challenge

### Limited animal evidence

Historical records cannot establish current animal use or welfare.

### Competing habitat needs

Planting and machinery may disturb cover, open habitat or breeding sites.

### Uncertain recovery

Satellite change alone does not establish habitat recovery speed.

## Tierra del Fuego approach

### Traceable environmental evidence

Combine CEMS mapping, Sentinel imagery and dated GBIF records.

### Explicit animal concerns

Inspect food, water, shelter, movement and breeding for six groups.

### Conditional options and review

Compare restoration pathways and record what needs field verification.

02. Approach

# Evidence and decision workflow

1. **Retrieve real evidence**

   CEMS footprint observed 25 June 2022. Satellite pair: 10 June / 5 July 2022.

2. **Compare matched windows**

   GBIF observations: June 2021 / June 2023, all groups, 5 km outside buffer.

3. **Interpret and document**

   AI draft, five values, conditional options and a PDF.

Uncertain locations: 143 / 204 records. Buffer is not a validated unaffected control.

## Actual demo results

Greenness (NDVI): 0.482 / 0.385
Spectral change (dNBR): 0.114
Inside footprint: 13 / 0 records
Nearby buffer: 35 / 5 records

**Historical records / no population inference**

# Why Design for Values?

From a broad ethical perspective to requirements we can test in Sierra de la Culebra

## WHY WE CHANGED METHOD

### Multispecies Justice

Established why affected animals should be considered in restoration decisions.

### Our practical constraint

A comprehensive justice assessment needs local species and welfare evidence we cannot yet verify.

### Design for Values

Turns the project's ethical priorities into norms, tool features and expert checks.

## FIVE VALUES → DESIGN IMPLICATIONS

**01 ANIMAL WELFARE** — PRIMARY OBJECTIVE

Flag likely effects on shelter, food, movement and breeding habitat.

**02 ECOLOGICAL RECOVERY** — SECONDARY OBJECTIVE

Compare habitat structure, natural regeneration and recovery suitability.

**03 TRANSPARENCY** — GOVERNING VALUE

Show evidence sources and assumptions behind each pathway.

**04 UNCERTAINTY AWARENESS** — GOVERNING VALUE

Label observed, inferred and unknown conditions; request field checks.

**05 ACCOUNTABILITY** — GOVERNING VALUE

Require qualified human review before any intervention is approved.

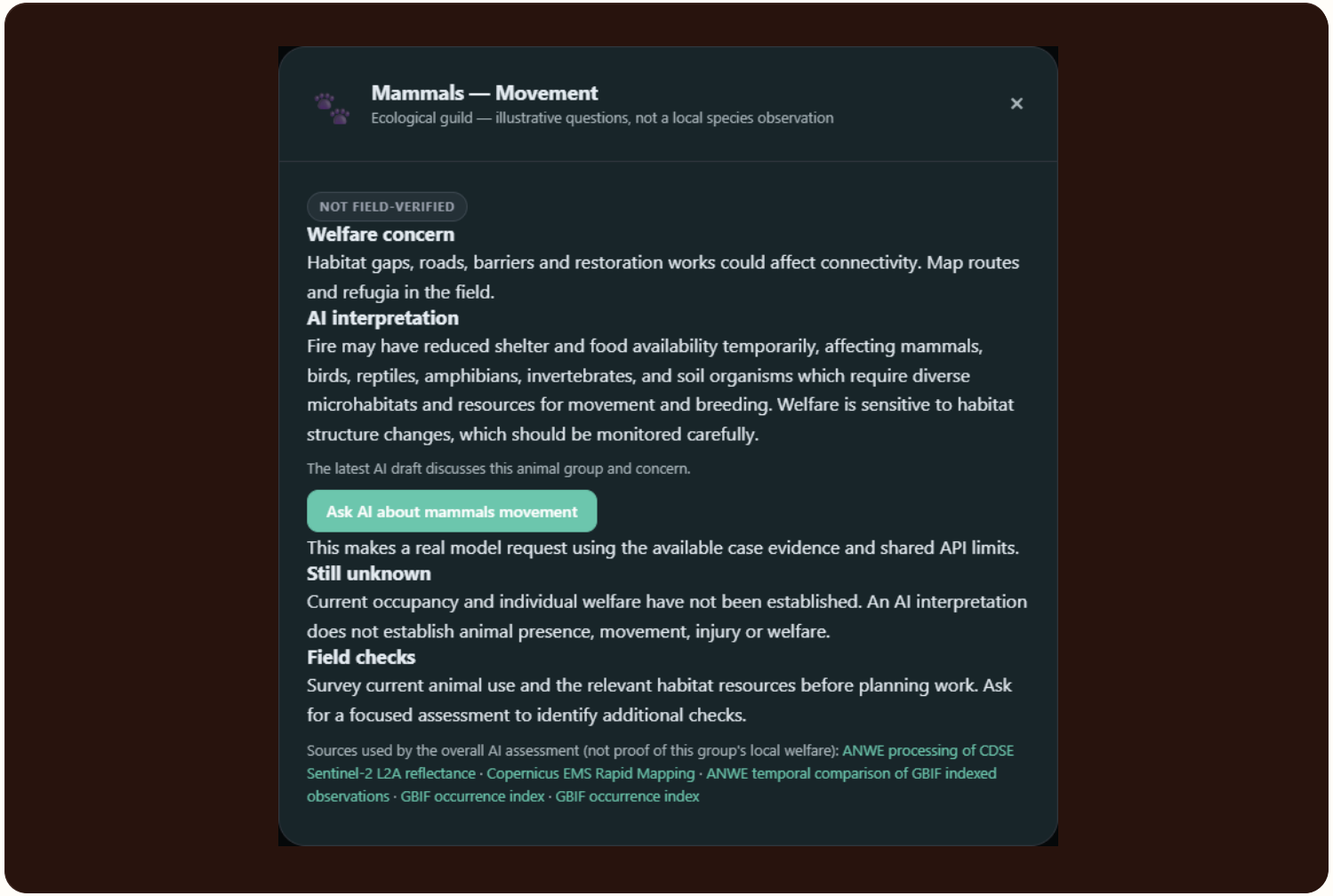
Initial values: project purpose + anticipated stakeholder needs + ecological evidence; expert validation still required.



Celermajer et al. (2021) · van de Poel (2013)

04. Interface

# Animal welfare and AI in the demo




Actual mammals movement popup / real session draft / no new model call for these slides

## Actual model response

gpt-4.1-mini considers the five values and three restoration options.

## Field questions remain visible

Each concern shows hypotheses, unknown conditions and checks. Current occupancy remains unverified.

# A report for restoration decisions

RESTORATION TEAMS

### Compare conditional options

Natural regeneration, assisted restoration and active restoration. Survey animal use and disturbance.

LAND MANAGERS

### Document the decision

28-page report with actual evidence, AI draft, full retrieved species lists, methods and glossary.

ECOLOGISTS & FIELD STAFF

### Verify conditions before action

Check refuges, connectivity, breeding, regeneration and erosion. Qualified review and permissions follow.